

Effective Utilization of Large-scale Unobserved Data in Recommendation Systems

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Abstract

Ranking models play an important role in industrial recommendation systems. However, most ranking models are trained only with the observed items but used to retrieve all items in the entire space, which may suffer from the sample selection bias and the exposure bias. Inspired by the entire space learning framework, we carry out detailed data analyses on large-scale unobserved items and find that they contain quite a few "potentially-positive" samples. In this paper, we propose an "Extract and Transfer" (EAT) framework, utilizing quantities of unobserved items and other domains' data to construct more training data for ranking models. Specifically, we first extract "potentially-positive" samples and negative ones according to their ranking scores from the unobserved data, and then design an Entire Space Transfer Learning (ESTL) model to transfer knowledge between observed and unobserved samples, instead of directly mixing them together to avoid negative transfer. Experiments on production data collected from Taobao validate the proposed method's superiority. Besides, we have deployed EAT on the Taobao recommendation system, obtaining 6.22% IPV (Item Page View) and 3.77% CTR improvement. The code is available at <https://github.com/Recommender1/EAT.git>¹.

CCS Concepts

• Information systems → Information retrieval; • Retrieval tasks and goals → Recommender systems.

*Both authors contributed equally to this research.

¹According to the company policy, we can only provide partial pseudo code.

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Keywords

Unobserved items, Recommender System, Transfer Learning, Multi-domain recommendation

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1 Introduction

Deep learning has been widely used in ranking models [4, 5, 29] and performs better than the traditional methods [3, 21, 29]. However, ranking models are faced with two challenges: 1) The Sample Selection Bias (SSB). Most ranking models are only trained with the observed items, but its serve space are all items returned by the previous stage, including a large number of unobserved items. The SSB makes the training data distribution deviate from test data distribution, meaning the training data only gives a skewed snapshot of user interest. It makes the ranking model overfit the observed items and has a negative effect on ranking models' performance [2]; 2) The Exposure Bias (EB). In real-world applications, users are only exposed to a small portion of items so the unobserved items are not always negative. To be specific, ambiguity arises in the interpretation of the unobserved items because the user is not aware of them. If ranking models can't distinguish real negative samples from the potentially positive ones, it results in significant bias, detrimentally affecting RS performance

To alleviate the SSB problem, many approaches have been proposed, which can be divided into two categories: auxiliary information [3, 28] and auxiliary domains [8, 14, 17]. For example, Ma et al. [16] uses ESMM to train CVR with all impression items to alleviate SSB and Data Sparsity (DS) problem. Chen et al. [3] proposes ESAM to transfer knowledge from displayed items to non-displayed items to alleviate the distribution inconsistency. Wen et al. [27] leverage all the impression samples across the entire space and employ more behaviors between click and purchase to address SSB problem. To deal with the EB problem, traditional methods treat all the

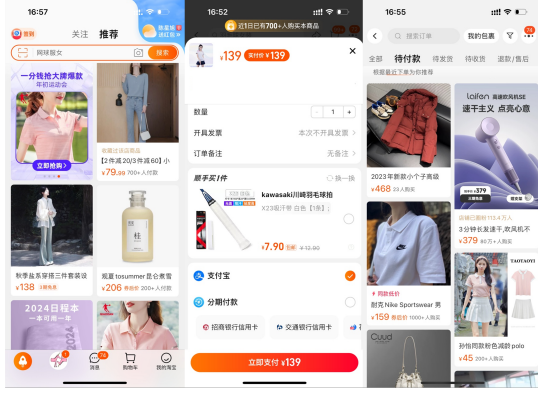


Figure 1: Multiple scenarios in Taobao recommendation: Guess You Like, Handbuy and Guess You Like after Purchase.

unobserved items as negative and specify their confidence levels [13, 18]. To further utilize behaviors in other scenes, Lyu et al. [14] propose an entire space learning framework to rebuild CVR on the entire space and explicitly introduce purchase samples from other domains, showing excellent performance. However, they only uses Pay_aToPay_g to predict the probability of sample bought in our scene of other scenes, which may has a negative effect on the performance of ranking models.

In real-world applications, CTR also suffers from SSB and EB. For example, Taobao recommendation CTR is only trained with observed items: clicked samples are labeled as positive and unclicked ones as negative. However, the observed items account for only 2% of the pre-ranking outputs, or even less. Ignoring the massive number of unobserved items may results in SSB. Moreover, real-world applications encompass multiple domains. For instance, Taobao, one of the largest e-commerce platforms in China, contains scenarios such as pre-purchase (Guess You Like), in-purchase (Handbuy) and post-purchase (Guess You Like after Purchase), among others, as shown in Fig. 1. Since different scenarios often have overlapping users and items, information sharing is beneficial for the CTR model in each domain, which can help a specific domain determine users' interest in items through interactions between users and items in other domains.

To solve the aforementioned problem in real-world RS, we follow the past works and propose EAT, a two-pipeline approach that utilizes quantities of unobserved items and data from the entire space to construct more training data for ranking models. Specifically, we first extract potentially-positive samples from the unobserved items when the items are clicked in other domains, following Lyu et al. [14]. Although these items are not displayed in the current scene, the interaction in the entire space indicates that the users may be interested in these items. For the negative samples, we adopt an adaptive sampling strategy: the higher the ranking score of an item, the greater the sampling probability of sampling it, which will be described in detail in section 3.1. Secondly, we propose ESTL, which is trained with both the observed and extracted samples. ESTL combines a multi-domain CTR model with a discriminative transfer network to transfer knowledge between observed and unobserved

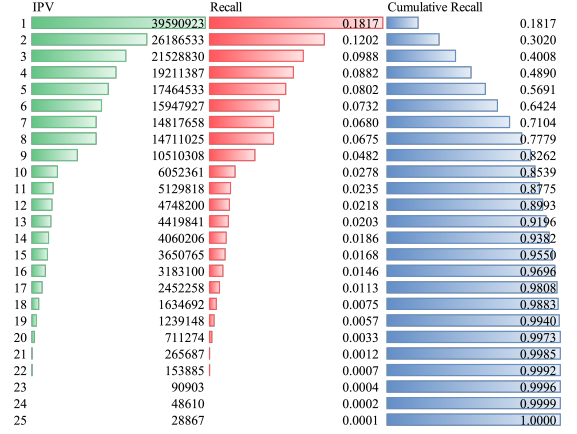


Figure 2: The distribution of "entire-space-positive" items in the items returned by ranking models. Recall is the recall value for the current bracket. Cumulative Recall is the sum of recall values up to the current bracket.

samples, enabling CTR learn more accurately. The motivation behind EAT will be introduced in Section 2. In summary, the main contributions of our work are:

- We conduct a detailed data analysis on the results returned by ranking models and uncover the value of large-scale unobserved items.
- We propose EAT, an approach that effectively utilizes large-scale unobserved items to improve the performance of ranking models by alleviating SSB and EB. Furthermore, EAT can be easily integrated into most ranking frameworks;
- We deploy our work in the CTR of Taobao recommendation, which serves hundreds of millions of people every day. Online A/B testing demonstrates the superiority of this work, and the details will be described in Section 4.3.

2 Analysis and Motivation

In this section, we want to construct a detailed data analysis on the trillions of real industrial data returned by ranking models and explain our motivation.

As the entire space learning framework suggests, items that are unobserved in the current domain but purchased in the entire space can contribute to improving the performance of CVR. Inspired by this insight, we seek to explore the value of the items unobserved in the current domain but clicked in other domains for the enhancement of CTR learning. For convenience, we refer to items unobserved in the current domain but clicked in other domains as "potentially-positive" items in this paper. Different from the real positive samples (exposed and clicked), these samples merely reflect a weak connection between users and items. However, they can also indicate users' preference to some degree, which we believe is valuable.

To conduct a detailed analysis and explore the real distribution of "entire-space-positive" items, including both the real positive and the potentially-positive ones, among the items returned by the model, we use the data collected from Taobao as an example

². We categorize trillions of items into 25 brackets based on their scores, arranged from high to low, and the first bracket represents the items exposed to users. Figure 2 displays the result.

Intuitively, among the items exposed to users, only **18.17%** are entire-space-positive items, while the remaining **81.83%** are among the unexposed items. Besides, by analyzing user behavior logs, we found that each day, after users leave the current scene, there are about **180** million clicks in other scenes occurring on the unobserved data, which is more than **4** times the number of clicks on observed items. This result is shocking to us: it suggests the large-scale unobserved items contain quite a few potentially-positive samples, which to a certain extent reflect the user's interests but are completely ignored by traditional CTR models.

In addition, ranking models are trained only with the observed ones but the serving space contains extensive unobserved items that include numerous potentially-positive samples. The difference in data distribution and characteristics between the training space and the serving space may lead the SSB and EB.

In conclusion, The motivations behind EAT are threefold:

- There are quantities of "potentially-positive" samples among the unobserved items. We want to rank these items higher than the negative ones by incorporating these samples into the training of the CTR model.
- Recall value decreases as the score decreases and the first half of the returned items contain the vast majority of the entire-space-positive samples. Hence, an adaptive sampling strategy, which tends to sample items with higher CTR scores, is necessary to sample negative items which are more challenging and important for CTR learning.
- Given the fact that the unobserved samples exhibits distinct data distributions and characteristics compared to the observed ones, we propose ESTL, which treats them as data from two different domains, rather than mixing them directly, to avoid negative transfer. In addition, ESTL also includes a discriminative network to implement instance transfer, enhancing the effectiveness of knowledge transfer.

3 Methodology

To alleviate the SSB and EB in CTR, we propose EAT, a two-step pipeline. The first step is to extract potentially-positive and negative samples from the unobserved items. The second step is to train ESTL with both original samples and the samples extracted in the first step.

3.1 Extract Samples from Unobserved Items

In this section, we will introduce the respective sampling strategy of positive samples and negative samples used in this paper.

The samples are divided into two parts: observed and unobserved ones. For the observed ones, we label the clicked items as positive and the un-clicked ones as negative, which is consistent with most existing methods. For the unobserved ones, following the analysis of Section 2, we extract the items that are unobserved but clicked in other scenarios as potentially-positive samples to alleviate ES: users

and items are overlapping across all scenarios, thus the behaviors in other domains also indicate users' preference for the items.

Simply extracting positive samples may lead to overfitting, while training with all negative samples is too time-consuming. Therefore, negative sampling is equally important. Recent works have shown that Hard Negative Sampling (HNS) can not only accelerate the convergence but also improve the model accuracy [6, 23]. In this paper, we sample negative samples by Softmax-based Sampling methods according to their prediction scores. The higher the item's score, the higher sampling probability it will be sampled. The negative sampling probability is defined in Eq. 1.

$$p_{neg_sample}^{Softmax}(i|c) = \frac{\exp(score_i)}{\sum_{j \in I_c^-} \exp(score_j)} \quad (1)$$

where $score_i$ represents the prediction score of unobserved item i , and I_c^- represents the set of all negative unobserved items. This sampling strategy is more inclined to sample highly-scored negative items, which are more informative.

3.2 Entire Space Transfer Learning Model

As there's a distribution difference between the samples extracted from unobserved and observed items, directly mixing these samples and train a CTR model is not rational and may lead to negative transfer. Inspired by multi-domain learning, this paper proposes an Entire Space Transfer Learning (ESTL) model that transfers knowledge between observed and unobserved items. The main architecture is depicted in Fig. 3.

Specifically, to avoid negative transfer, we treat observed and unobserved items as samples from "two domains" and design separate networks for them to learn specific knowledge. We will elaborate on ESTL in this section, including Shared Experts, Private Experts and the Discriminator Network module.

3.2.1 Shared Experts: Learn the Common Knowledge. Although observed and unobserved items have different data characteristics, they are returned by the same ranking model, so they naturally have some similarities. Therefore, knowledge sharing between them can help improve the performance of ranking models.

Inspired by Multi-gate Mixture-of-Experts (MMOE), we design three shared experts to capture the commonalities, and then adopt two gating networks to produce the "common-knowledge" for observed and unobserved items respectively, as calculated by Eq. 2.

$$CommonKnow_i = \sum_{m=0}^M Sexpert_m gate_i^m \quad (2)$$

where $CommonKnow_i$ means the common knowledge learned for $domain_i$, $Sexpert_m$ is m -th shared expert's output, and $gate$ represents the gating function and $gate_i$ is used to dynamically balance these shared experts' output for i -th domain. "Shared Experts" are shared by all domains, so its training samples are all items, including observed and unobserved ones. "Gating network" is trained to learn the weights of these experts for the specific domain, thus it is trained with the domain samples: observed or unobserved ones.

3.2.2 Private Expert: Learn the Specific Knowledge. As Tang et al. [25] suggest, explicitly separating shared and task-specific experts is important to address negative transfer. In this work, both observed

²The data used for the data analysis has been anonymized. It is not the raw data collected from Taobao, but the data distribution and trends remain consistent.

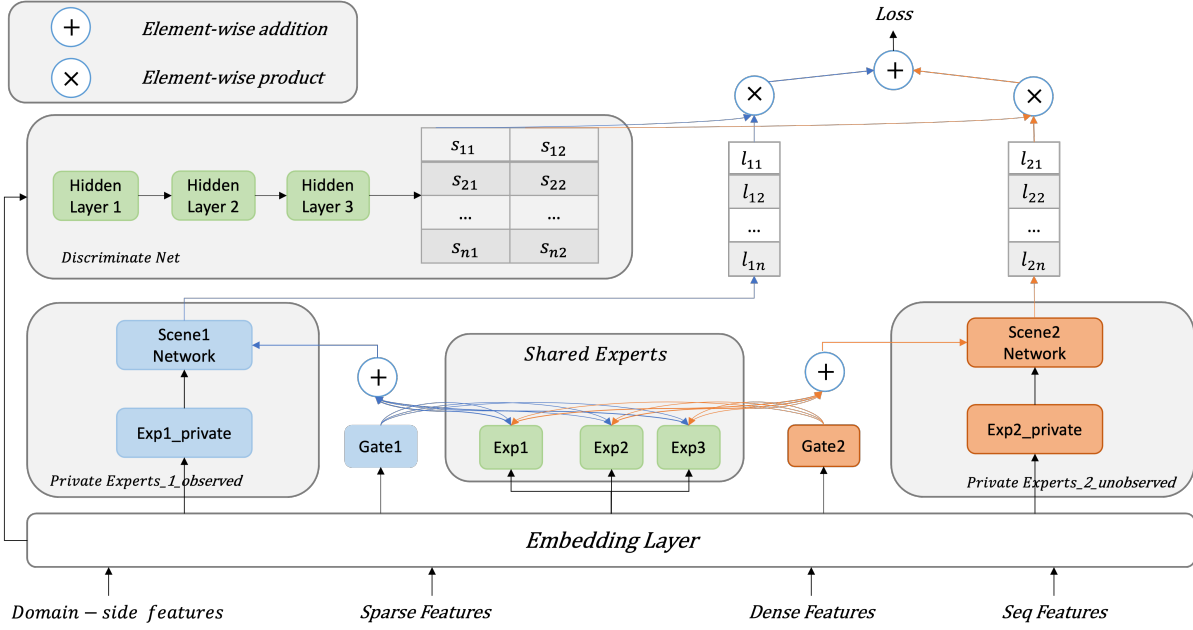


Figure 3: Structure of ESTL this paper proposes. The number 1 represents the observed samples network (blue), including gate1, private_expert1 and Scene1; The number 2 represents the unexposed samples network (orange), including gate2, private_expert2 and Scene2; and the green section represents the common parts, including the shared experts and the discriminative model.

and unobserved items have a private expert, which is trained to learn the specific knowledge. Each private expert consists of three hidden layers, and is trained only with the observed samples or the unobserved items, similarly to the gate network. Then we concatenate the output of the private network $Pnet_i$ with $CommonKnow_i$, and feed them into a unified network $Scene_Network_i$, which combines common knowledge with specific knowledge. Finally, the output is formalized as Eq. 3

$$SceneNet_i = Scene_Network_i([CommonKnow_i, Pnet_i]) \quad (3)$$

where $Scene_Network$ also consists of three hidden layers, and its output can represent the interaction between the user and the item learned by both shared experts and the private expert.

3.2.3 Discriminator Network: Instance Selection. To further facilitate knowledge transfer and mitigate negative transfer, a discriminator network is introduced. The discriminator module, an instance-level method, selects instances from the source domain that have close distribution as the target domain.

Different from the domain adaptation (DA), the discriminator module is a cross-domain-instance-transfer method. It can transfer instances from observed to unobserved samples, as well as from unobserved to the observed ones. Specifically, we pre-train a Multi-class Classification Network to learn the probability of a sample belonging to a domain. The k -th sample from j -th domains trains the j -th private expert regardless. If the probability p_i^k of the k -th sample being classified as the i -th domain is greater than the threshold ϵ , it will also train the i -th private expert, as

shown in Eq. 4.

$$sample_j^k : \begin{cases} \text{train } j\text{-th private expert} & \text{regardless} \\ \text{train } i\text{-th private expert} & \text{if } p_i^k > \epsilon \end{cases} \quad (4)$$

The structure of the discriminator module is based on three-layers feed-forward network. It is worth noting that since the task of this module is quite different from the task of the main networks' task, the gradient of discriminator network does not update the bottom embedding layer.

4 Experiments

In this section, we conduct extensive experiments to evaluate the efficacy of EAT, with the purpose of answering the following key questions:

- **RQ1:** What metric should we use to evaluate the offline performance of the proposed method to ensure consistency with online performance?
- **RQ2:** How well does the proposed method perform compared with the commonly used methods in industry? Both the offline evaluation results and online A/B test results need to be presented in detail.
- **RQ3:** What is the influence of major components (including the sampling strategy and discriminator network), and the effect of the hyper-parameter (the sampling probability) on recommendation results?

4.1 Experimental Settings

Datasets. EAT needs all items' features and scores, including large-scale unobserved items, return by the CTR model. Due to the lack

of suitable public datasets, we collect an industrial dataset from the traffic logs of Taobao App³. We construct an offline dataset based on users' behaviors in ten days. The first nine days' data are used as training set, and the last day's data as test set. The dataset contains trillions of examples, making the training time is too long to satisfy the updating of industrial recommendation systems. Therefore, we sample the data according to the strategy introduced in the Section 3.1 and construct a dataset, which consists of hundreds of billions samples. The potentially-positive samples are labeled as 1 as well ⁴. **Baselines.** To demonstrate the effectiveness of EAT, we compare it with the following methods:

- YoutubeNet [5]. The structure of "embedding&MLP" is widely adopted in real-world recommendation system, and the same goes for Taobao. Thus we use it as our base model.
- Mixing observed and unobserved samples directly. This method simply mixes all samples together without distinguishing between observed and unobserved ones.
- Random sampling. We adopt a Softmax-based Sampling Method to attach more importance to the highly-scored negative samples. By comparing our method with random sampling method, we can demonstrate its effectiveness.
- MMoE [15]. MMoE is a machine learning model framework used for multi-task learning problems, which captures the commonalities and differences among various tasks through the use of different "expert" networks. It is a commonly used framework in RS.
- PLE [25]. PLE is a novel multi-task learning architecture that progressively extracts and integrates features across related tasks, which is widely used in RS.

4.2 Offline Evaluation Metrics(RQ1)

Area Under Curve (AUC) is traditionally used for offline CTR evaluation. However, EAT alters the distribution of the sample space and the impression items will be different. Thus, AUC measuring only the ordering quality of observed items is not rational.

After analysis and comparison, we adopt the following metric to evaluate the quality of the recommended item list: weighted-AUC, which can make offline evaluation be consistent with online performance. It means that if one model outperforms others in offline evaluation, it is expected to surpass other models in online performance as well.

Weighted-AUC calculates the AUC for weighted samples. Our method aims to rank potentially-positive items that are ranked in the middle or at the bottom higher so that they can be exposed to the users. Accordingly, both AUC for observed items and AUC for unobserved items need to be considered. Since positive observed items are real positive ones (exposed and clicked), but the positive unobserved items are potentially-positive (unexposed but clicked in other domains), they cannot be weighted equally. Therefore, we should assign different weights to them respectively. Obviously, the observed samples are more valuable because they reflect the real interaction between users and items in the current domain, so we assign a weighting coefficient of 2 to the exposed items and 1

Table 1: Offline results. * indicates the performance difference against the second-best result is statistically significant ($p < 0.05$). To ensure fairness, aside from w/o unpv and ESTL+random, all experiments employed the adaptive-sampling strategy. It is noteworthy that in practical RS, an AUC improvement in the thousandths is significant.

	weighted AUC	Δ
w/o unpv	0.6353	-
mixed	0.5754	-0.0599
ESTL+rand	0.6389	+0.0036
MMoE	0.6388	+0.0035
PLE	0.6396	+0.0043
ESTL+adaptive	0.6409*	+0.0056

to the unexposed items. These weight coefficients are determined through extensive experiments and empirical evidence, and how to better integrate these two terms could be a direction for the future research.

4.3 Recommendation Performance (RQ2)

We evaluate all the approaches using the aforementioned Taobao production dataset⁵. In this section, we present the results of our experiments, including both offline and online results.

4.3.1 Offline Evaluation. We report the offline results of the methods in Table 1. EAT outperforms all the baseline models. For brevity, we refer to the first model trained without unobserved samples, as the base model. ESTL+adaptive sampling strategy is EAT.

Compared with the base model trained only with observed items, mixing the unobserved and the observed items directly results in a significant performance decrease, with -0.0599 weighted-AUC decrease. The decrease illustrates that treating observed and unobserved items equally is irrational, which may lead to negative transfer, greatly compromising the model performance.

Compared with the base model, EAT achieves a $+0.0056$ improvement in weighted-AUC. Meanwhile, compared to the commonly used multi-scenario learning frameworks, MMoE and PLE, there is an improvement of 0.0021 and 0.0013 respectively, demonstrating the effectiveness of the structure we propose, which can efficiently transfer the knowledge between observed and the unobserved items.

There's a large quantity of unobserved items, and it is impossible to train CTR with all the samples because such a long training period is unacceptable. As section 3.1 introduces, we adopt the Softmax-based sampling strategy. To validate the effectiveness of the strategy, we compare it with the random sampling strategy. ETA+adaptive sampling strategy we adopt achieves absolute weighted-AUC gain of 0.002. The comparison result shows that attaching more importance to the hard negative samples of the unobserved samples is greater, has a positive effect on CTR performance.

³<https://www.taobao.com/>

⁴They also reflect the weak link between users and items. But we don't treat them equally in ESTL.

⁵There is no suitable public dataset. We can only use an industrial dataset from Taobao for experiments and are unable to release the dataset according to the company policy. However, we have anonymized the data and will release the dataset upon acceptance.

Table 2: The online performance of all the methods deployed in Taobao recommendation. The best results are highlighted in bold.

	IPV	CTR
w/o unpv	-	-
mixed	-7.72%	-8.22%
ESTL+rand	+4.46%	+2.40%
MMoE	+4.16%	+2.20%
PLE	+5.19%	+2.91%
ESTL+adaptive	+6.22%	3.77%

4.3.2 Online Evaluation. We deploy these methods to Taobao for personalized recommendation and conduct careful online A/B testings to evaluate the online performance of EAT. To compare fairly, except CTR model which is trained with different methods, other variables, including CVR, user pool, item pool, etc., maintain consistent. We distribute users into multiple buckets, with each model to one of them.

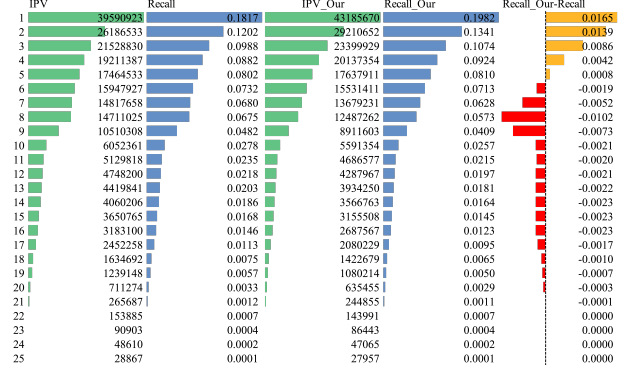
As shown in Table 2, EAT performs best in online experiments, outperforming all baseline models. Compared with the base model, ESTL+adaptive sampling strategy increases Item Page View (IPV) and CTR by 6.22% and 3.77% respectively⁶. The results further confirm that EAT we propose is effective to capture users' interests from the large-scale unobserved items and improve the performance of CTR. In addition, compared with the experiment directly mixing all samples, ESTL+random can enhance the performance of CTR significantly. These online results are consistent with the offline evaluation and further validates the rationality of the offline metrics.

4.3.3 Changes in entire-space-positive items distribution. As shown in Table 1, the weighted-AUC of ESTL+adaptive sampling strategy surpasses the base model, which illustrates the value of large-scale unobserved items. Moreover, it demonstrates that training a CTR model with unexposed data can reduce the discrepancy between training and testing datasets, alleviating the SSB and enhance its performance across the entire test space. Extracting potentially-positive samples with the aid of other domains' data, instead of treating all unobserved items negative helps mitigate the EB.

To more intuitively demonstrate the effectiveness of our method, we show the difference in the distribution of entire-space-positive items returned by the base model and ETA in Fig. 4.

Recall of the top five brackets increases, with the first bracket exposed to users showing the largest increase; recall of the mid-tail and tail brackets decreases and the middle brackets decrease distinctly. This reveals that EAT tends to rank items that were originally placed in the mid-tail or tail but clicked in the entire space higher, moving them to the top positions to give them the opportunity to be exposed to the users.

The results illustrate that EAT has a better ability to score the items in the entire space, That is, ESTL paying attention to the unobserved items as well may capture the users' interests better

**Figure 4: The distribution of "entire-space-positive" items in the items returned by the base model and ESTL we proposed respectively. Recall is the recall value for the current bracket. Recall_Our - Recall represents the delta in recall for each bracket of ESTL compared with the base model.****Table 3: Offline performance comparison between ESTL and ESTL-dm.**

	weighted AUC	Δ
ESTL	0.6409	-
ESTL-dm	0.6396	-0.0013

by transferring knowledge between the items observed and unobserved to alleviate SSB and EB, which is highly consistent with our expectations.

4.4 Analysis and Ablation (RQ3)

To study the effects of the major components and hyper-parameters of EAT, we conduct additional experiments.

4.4.1 Ability of the Discriminator Module. In this section, we demonstrate the importance of the discriminator module. We denote ETSL without the the discriminator module as "ESTL-dm". As shown in Table 3, "ESTL-dm" falls behind, with -0.0013 decrease in weighted-AUC. These results further demonstrate the effectiveness of the discriminator module, which enhances knowledge transfer between observed items and unobserved items through instance transfer.

4.4.2 Hyper-parameter Study. To study the effect of the sampling probability p , we vary the value as $p \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$. We report weighted-AUC with different p in Figure 5. With $p = 0$, the extraction progress is not applied, and we train the CTR model only with observed items, which is consistent with the baseline model. With $p = 1.0$, all the unobserved items are extracted. The results illustrate that p makes a great difference to the model performance. The model achieves best performance when $p = 0.5$.

4.5 Engineering Optimization Strategies.

To deploy EAT in Taobao recommendation system, we follow the online and offline strategies.

⁶It is worth noting that a 6.22% increase in IPV is significant in Taobao recommendation

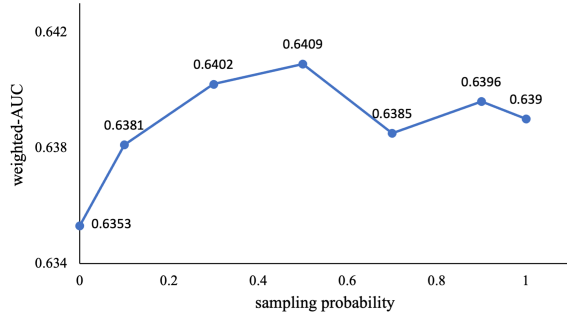


Figure 5: Experimental results with different negative sampling probability p .

Online Serving Strategy. During the offline training, we train two separate experts for observed items and the unobserved items. However, when serving online, we cannot determine in advance whether an item will be exposed. Therefore, all the items in the entire space are scored by the observed items network. The online performance confirms the feasibility of our method. Exploring better ways to utilize two scenario-specific networks is a topic for future research.

Feature construction. In RS, user behavioral features, particularly the sequence of behaviors, are constantly changing in real-time. It is unreasonable to construct user features using offline associations, as this may result in significant discrepancies between the features during training and online serving. Therefore, we employ full-tracking, which captures the user features and item features at the moment of the user’s request as features of training samples.

Offline Training Strategy Our samples contain both observed items and unobserved items, which vary greatly in quantity. We adjust the proportion of each in every batch according to their quantities. This prevents that a surplus of unobserved samples are left when the observed samples have been consumed completely, which could cause catastrophic forgetting.

5 Related Work

Our work builds on CTR prediction, contains the application of transfer learning, and is closely related to multi-domain learning. In this section, we give a brief introduction to multi-domain learning and transfer learning respectively.

5.1 Multi-domain Learning

In real-world application, training samples are collected from multiple domains, training one model for each domain is time-consuming. Thus multi-domain learning has been widely applied in recommendation system [1, 12, 22], which aims to transfer information between domains to improve recommendation quality. The Collective Matrix Factorization (CMF) [24] model is proposed to learn shared representations of users of items but jointly factorizes matrices from different domains. Cross-Domain Collaborative Filtering (CDCF) [11] is proposed to transfer knowledge from auxiliary domains to a target domain to address data sparsity and cold-start issues in collaborative filtering. Neural network-based models are applied

to multi-domain learning due to the rapid development of deep learning. Multi-gate Mixture-of-Experts (MMoE) model [15] uses gating networks to combine multiple expert networks, allowing the model to capture both domain-specific and shared knowledge. Graph Neural Networks (GNNs) [7] are used to address the multi-domain recommendation, which enables the model to learn more generalized user representations. Star Topology Adaptive Recommender (STAR) [22], which consists of shared centered parameters and domain-specific parameters to solve the multi-domain CTR prediction. Parameter and Embedding Personalized Network (PEP-Net) [1] learns the heterogeneous relationships between multiple domains and multiple tasks to accurately capture users’ interests. Progressive Layered Extraction (PLE) [25] separates task-sharing and task-specific parameters, and proposes the innovative progressive routing manner to achieve more efficient information sharing. Therefore, multi-domain learning improves the performance of the recommendation system significantly by achieving effective knowledge sharing across different domains.

5.2 Transfer Learning (TL)

Multi-domain learning transfers knowledge between multiple domain and has been introduced in section 5.1. To reduce negative transfer, the instance-based method is widely used, which selects a part of training instances from the source domain which share a similar data distribution with the target domain [19, 26]. Traditional methods need to measure the similarity between the source and target domain, and adjust the weight values for the source domain samples. For instance, Bayesian Optimization is used to select data for transfer learning [20]. However, these methods are not suitable for deep learning models because they are challenging to be integrated into an end-to-end learning framework [10].

The Reinforced Transfer Learning (RTL) [19] method trains a data selector according to the reward returned by the TL model. A general Minimax Game based model for selective Transfer Learning (MGTL) [26], containing a selector, containing a selector, a discriminator and a TL module, is proposed to help select useful source data for transferring. Adaptive Clustering Transfer Learning (ACTL) [9], which trains the cluster-based instance selector with the TL model together, learns to select instance and improves transferability.

6 Conclusion

In this paper, we propose a two-step approach, namely EAT, to improve the performance of ranking models. Specifically, we first extract potentially-positive samples from large-scale unobserved items, and then train ESTL with these samples to transfer knowledge between observed and the unobserved items while avoiding the negative transfer. It is worth noting that EAT is a general framework that can be easily integrated into existing ranking models. We have successfully implemented our framework into Taobao recommendation system. Both offline and online experiments demonstrate that our method achieves significant improvements. For future work, we want to investigate how to combine the observed and the unobserved networks better to score the items across the entire space during online serving.

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